

A New Design Method of LCL Filter for Single Phase Grid Connected Power Converter

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Abstract—This paper proposes a new design method of LCL filter for a grid connected single phase inverter to improve reduction of switching ripple current. LCL filter is designed from the viewpoint of Total Harmonic Distortion(THD), reactive power and minimization of magnetic materials. The converter side filter inductor L_1 depends on a ripple ratio of the switching frequency component of the converter side current to the rated current, which varies by the modulation index and PWM method of the inverter. Filter capacitor limits the amount of reactive power absorbed from the grid. The grid side filter inductor L_2 depends on an attenuation ratio of the switching frequency component of the converter side current and the grid side current. The validity of filter design is verified by simulation of current control stability and current ripple characteristics by applying designed LCL filter to the single phase inverter, which can be applied to a railway application.

Keywords—LC filter, single phase inverter, ripple ratio, attenuation ratio

I. INTRODUCTION

In recent years, interests in the renewable energy such as photovoltaic and wind power and the integrated energy storage systems have been rapidly increasing as a countermeasure for temperature change caused by global warming. In order to link with these systems, a power electronic converter for the renewable sources is essentially required. The power converter applied for the grid connection produces the harmonic components of the output current flowing into the grid due to the semiconductor device performing the high frequency switching operation. Since such current ripple components cannot be suppressed by the control method, a physical output filter is required to improve power quality. The total harmonic distortion (THD) of the output current should be limited to be less than 5% in order to meet the grid code and improve the quality of the power flowing into the power system [1]. Among the output filter topology, L filter is the simplest structure but has disadvantages such that it is expensive and has a large volume [2]. Since LCL filter is able to reduce the total filter volume and improve the switching ripple reduction characteristic compared to L filter, researches on LCL filter are actively under way [3],[4]. In contrast to these advantages, design process of LCL filters is complicated, and proper design process is required to improve the performance of LCL filter [5]. Although the higher order LCL filter provides 60 dB per decade attenuation for the PWM carrier and side band voltage harmonic, possible instability of the current control is caused by the LCL filter and so, many different damping methods have been discussed [6][7][8].

In this paper, a relationship between the filter inductance and the switching ripple ratio in the grid connected voltage fed PWM inverter with a single phase full bridge structure is

analyzed, and a new design method of the LCL filter satisfying the given THD requirement is proposed. The validity of the proposed design method is verified through simulation.

II. ANALYSIS OF LCL FILTER

A. Single phase grid-connected converter

Fig. 1 shows a configuration of the single-phase grid-connected PWM converter system, which consists of a full bridge inverter, DC link capacitors, LCL filter and low voltage single-phase grid. In Fig. 1 (a), V_{DC} represents a DC source, and it can be photovoltaic cells for the photovoltaic power plant, a rectified DC for the wind power plant, battery cells for the battery energy storage system, and DC capacitor bank for back-to-back converter. The back-to-back converter is used for a universal power converter topology in the industry, and Fig. 1 (b) shows a single phase three level back-to-back converter for the railway application, which consists of Scott transformer, converter M connected to the main single phase, converter T connected to the teaser single phase, DC capacitor banks and LCL filters. The output current of the converter is injected into the grid and harmonic components of the output current is reduced through the LCL filter.

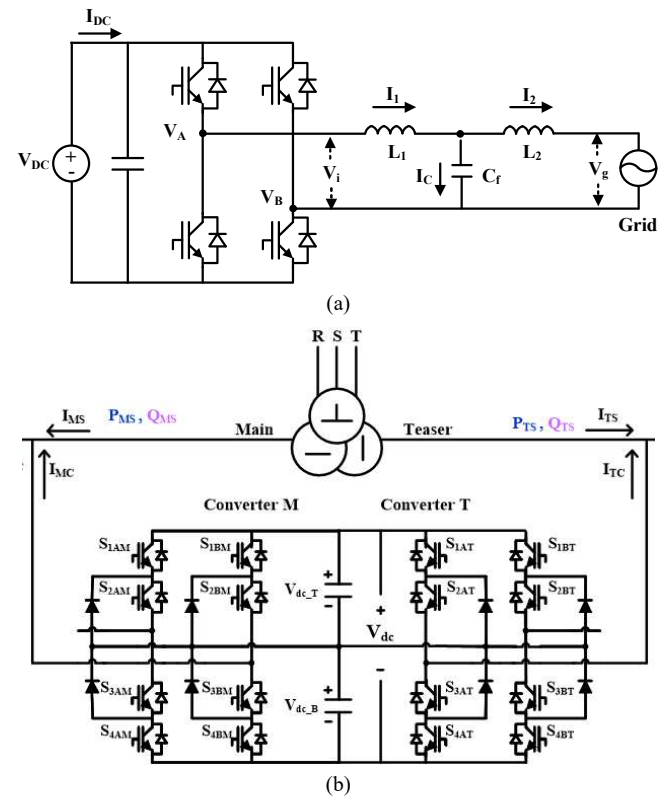


Fig. 1. (a) Circuit of single phase inverter and LCL filter. (b) back-to-back converter for railway application.

B. THD of L_1

It is ideal to feed only the current component corresponding to the rated frequency into the grid, but the output current necessarily includes the current ripple components corresponding to the switching frequency and side bands due to the switching characteristic of the PWM converter. Ratio of the switching frequency component to the fundamental frequency component of the converter side output current I_1 is represented as total harmonic distortion (THD) of the converter side output current THD_{i1} as shown in (1). I_r is the effective value of the switching ripple component and I_{rate} is the effective value of the rated current, where the smaller the current ripple rate, the better the power quality.

$$THD_{i1} = \frac{I_r}{I_{rate}} \times 100[\%] \quad (1)$$

Equation (2) is the peak-to-peak value of the ripple current flowing through the inductor L_1 with the single-phase inverter and unipolar PWM method applied, and equation (3) shows the ripple current effective value [9].

$$\Delta i_{pp} = \frac{V_{DC}T_s}{2L_1} (1 - m_a \sin(\omega t)) \times m_a \sin(\omega t) \quad (2)$$

$$I_r = \frac{V_{DC}T_s}{4L_1} \sqrt{\frac{2m_a^2}{3\pi} \left[\frac{\pi}{4} \left(1 + \frac{3}{4}m_a^2 \right) - \frac{4}{3}m_a \right]} \quad (3)$$

Fig. 2 shows the peak-to-peak value of the ripple current, and the magnitude is varied depending on the modulation index (m_a) and the phase of the current such as equation (2).

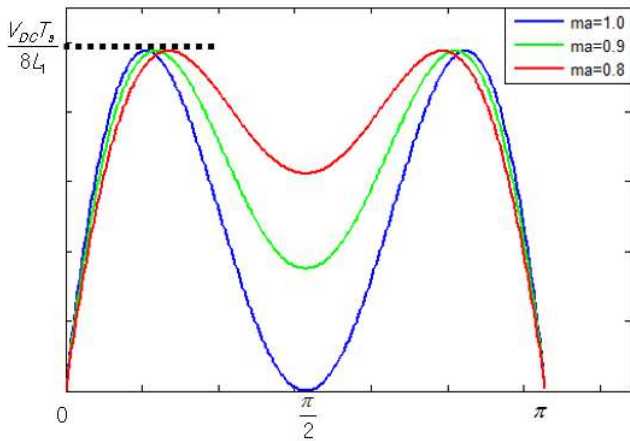


Fig. 2. The magnitude distribution of Δi_{pp} according to m_a

Fig. 3 shows the simulated result of the ripple current when modulation index m_a is 0.7 and it can be seen that an envelope magnitude of the ripple current is well matched with the curve in Fig. 2.

THD_{i1} is derived as equation (4).

$$THD_{i1} = \sqrt{\frac{\pi}{3} \left[\frac{\pi}{4} \left(1 + \frac{3}{4}m_a^2 \right) - \frac{4}{3}m_a \right]} \cdot \frac{T_{sw}}{T} \cdot \frac{L_b}{L_1} \quad (4)$$

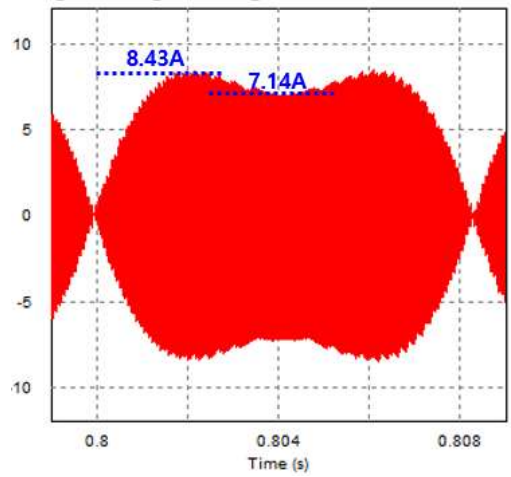


Fig. 3. Waveform of the ripple current ($m_a=0.7$)

C. Current Ripple Attenuation A

Current Ripple Attenuation, A is defined as the ratio of the inductor ripple current at the converter side to the inductor ripple current at the grid side. Fig. 4 shows the single-phase LCL filter circuit at the h -order harmonic.

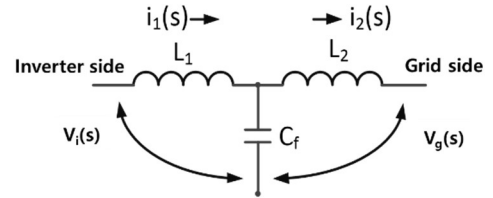


Fig. 4. Single-phase LCL filter circuit at h -order harmonic

From Fig. 4, the transfer function from inverter output voltage V_i to grid side current I_2 and one from inverter output voltage V_i to converter side current I_1 are derived as (5) and (6), respectively. By using (7) and (8), (5) and (6) can be more arranged into (9) and (10). Current Ripple Attenuation is derived by dividing (5) by (6) and shown in (11).

$$\frac{i_2(s)}{v_i(s)} = \frac{1}{s^3 L_1 L_2 C + s(L_1 + L_2)} \quad (5)$$

$$\frac{i_1(s)}{v_i(s)} = \frac{s^2 L_2 C + 1}{s^3 L_1 L_2 C + s(L_1 + L_2)} \quad (6)$$

$$\omega_{res}^2 = \frac{L_1 + L_2}{L_1 L_2 C} \quad (7)$$

$$Z_{LC}^2 = \frac{1}{L_2 C} \quad (8)$$

$$\frac{i_2(s)}{v_i(s)} = \frac{1}{s L_1} \frac{Z_{LC}^2}{s^2 + \omega_{res}^2} \quad (9)$$

$$\frac{i_1(s)}{v_i(s)} = \frac{1}{s L_1} \frac{s^2 + Z_{LC}^2}{s^2 + \omega_{res}^2} \quad (10)$$

$$A = \frac{i_2(s)}{i_1(s)} = \frac{1}{s^2 L_2 C + 1} \quad (11)$$

D. THD of L_2

THD_{i2} represents total harmonic distortion (THD) at the grid side current I_2 and can be expressed as (12).

$$THD_{i2} = THD_{i1} \times A \\ = \frac{T_{sw}}{T} \cdot \frac{2L_b}{L_T} \cdot \frac{2}{\omega_s^2 L_T C - 2} \cdot \sqrt{\frac{\pi}{3} \left[\frac{\pi}{4} (1 + \frac{3}{4} m_a^2) - \frac{4}{3} m_a \right]} \quad (12)$$

E. Inductance ratio $r (=L_2/L_1)$

The resonance frequency of the LCL filter is calculated as (13).

$$\omega_{res}^2 = \frac{L_1 + L_2}{L_1 L_2 C} \quad (13)$$

By defining an inductance ratio $r = L_2 / L_1$, (13) is rearranged to (14).

$$L_T C = \frac{k^2}{4\pi^2 f_{sw}^2} \frac{(1+r)^2}{r} \quad (14)$$

By differentiating (14), minimum value of $L_T C$ can be obtained with $r = 1$. Relationship between $L_T C$ and r is shown in Fig. 5.

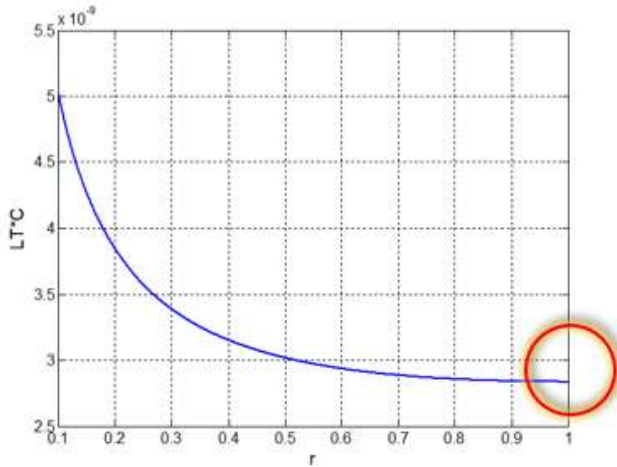


Fig. 5. $L_T C$ in terms of r

F. Filter capacitance C_f

Filter capacitance C_f is defined as product of the base capacitance C_b of the converter and the reactive power absorption rate x , where the base capacitance at the rated condition is given by (15).

$$C_b = \frac{1}{2\pi f Z_b} \quad (15)$$

Design value of the filter capacitor considering the reactive power absorption rate x is given by (16).

$$C_f = x C_b \quad (16)$$

III. DESIGN PROCEDURE OF LCL FILTER

When designing an LCL filter, required THD_{i2} , THD of the grid side current is determined at the first. Then L_T is calculated by using (12). Since the inductance ratio r is set to 1, value of L_1 and L_2 can be calculated. Filter capacitance C_f is calculated by using (16) with the reactive power absorption rate x . The design procedure is summarized as the below five steps.

Step I: System parameter selection such as system capacity (P), system voltage (V), DC link voltage (V_{DC}), system frequency (f_n), switching frequency (f_{sw}).

Step II: Determine THD_{i2} of the grid side current and $r = 1$ is used as the inductance ratio for minimum use of magnetic materials.

Step III: Calculate capacitor filter value by using the reactive power absorption rate x ($C_f = x \cdot C_b$). In general, the reactive power absorption rate x is around 2 ~ 5%.

Step IV: Calculate L_T by using THD_{i2} .

Step V: Calculate the resonant frequency (f_{res}) and check the resonant frequency ranges from $f_{sw}/6$ to $f_{sw}/2$ for stable operation. It should be noted that the grid side current feedback is employed for current control in this paper.

The following are design example and LCL filter is designed by employing the above design procedure.

- System capacity $P = 10\text{kW}$, grid voltage $V = 220\text{V}$, DC link voltage $V_{DC} = 450\text{V}$, grid frequency $f_n = 60\text{Hz}$, switching frequency $f_{sw} = 16.8\text{kHz}$
- THD_{i2} is limited to less than 5% in general and 0.3% is used in this design. From an economic point of view, r is chosen as 1
- Reactive power absorption rate x is selected as 0.03. With $C_f = C_b \cdot x$, value of filter capacitance C_f is determined as $15\mu\text{F}$.
- L_T value is selected as $200\mu\text{H}$ by using (12). L_1 is selected as $100\mu\text{H}$, L_2 is selected as $100\mu\text{H}$, with $r = 1$
- Value of $L_T \cdot C$ and resonance frequency f_{res} are calculated and f_{res} is ranging from $f_{sw}/6$ to $f_{sw}/2$ is confirmed.

Table I shows a calculation sheet for information.

TABLE I. CALCULATION SHEET

Parameter	Symbol	Value	Unit
Rated output power	P	10	kW
Grid voltage	V_g	220	V
DC link voltage	V_{DC}	450	V
Grid frequency	f_n	60	Hz
Switching frequency	f_{sw}	16800	Hz
Reactive power absorption rate	x	0.03	
Target THD_{i2}	THD_{i2}	0.003	
L_2/L_1	r	1	

Modulation index	m_a	0.7	
Grid period	T	16.7	ms
Switching period	T_s	60	us
base impedance	Z_b	4.84	Ω
base inductance	L_b	12.8	mH
bas capacitance	C_b	550	uF
Ripple Factor	RF	0.19	
Current Ripple Attenuation	A	0.016	
Total inductance	L_T	200	uH
Inverter side inductance	L_1	100	uH
Capacitance	C_f	15	uF
grid side inductance	L_2	100	uH
resonance frequency	f_{res}	5.9	kHz

IV. DESIGN VERIFICATION

Fig. 6 shows a circuit diagram of the single-phase grid-connected converter for railway application and Table II shows simulation parameters.

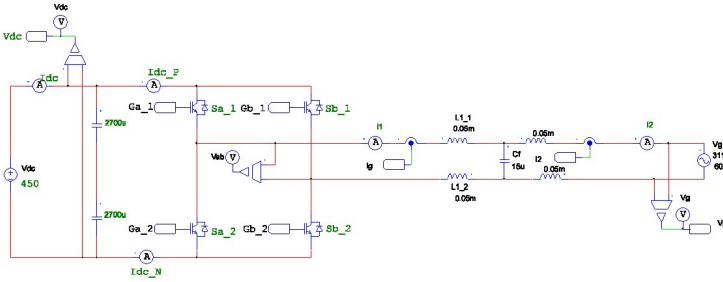


Fig. 6. Circuit of single-phase converter

TABLE II. SIMULATION PARAMETERS

Parameter	symbol	value
DC Voltage	V_{DC}	500V
Switching Frequency	f_{sw}	10kHz
L_1, L_2 filter	L	0.1mH
DC link capacitor	C_{DC}	375uF
Grid voltage	V_g	220V
Grid frequency	f_n	60Hz

The designed LCL filter at section III is applied to the simulation circuit and the performance of the designed LCL filter is verified through simulation. Fig. 7 shows waveform of converter side current I_1 (red colored) and grid side current I_2 (blue colored) from the start-up to the steady state. Current control is stable with the designed small LCL filters and a conventional current control method.

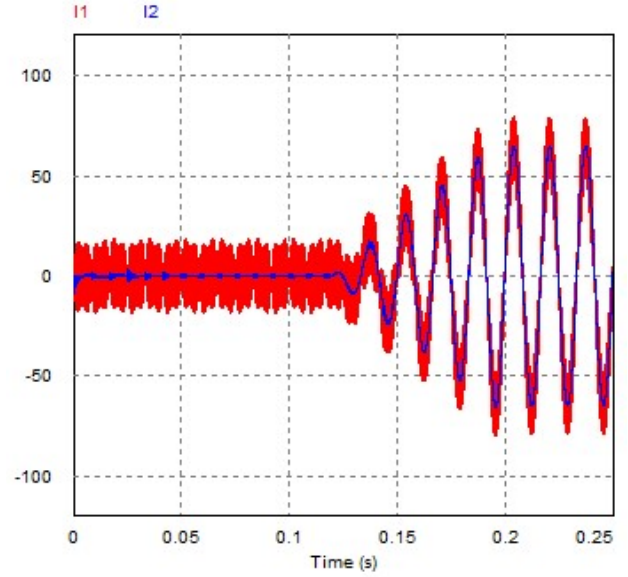


Fig. 7. Waveform of I_1 and I_2

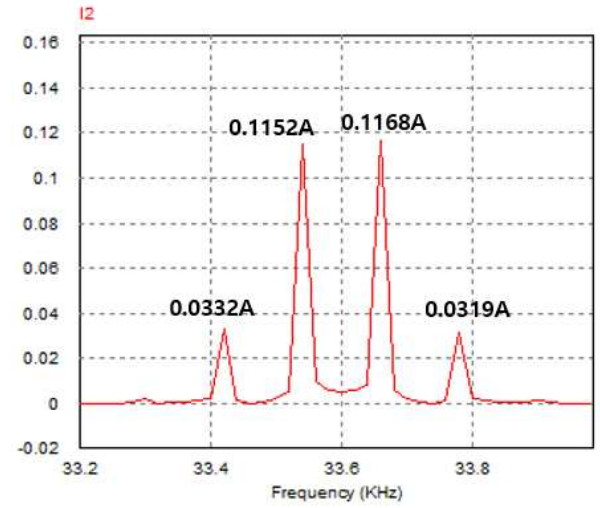


Fig. 8. Spectrum analysis (FFT) of I_2 .

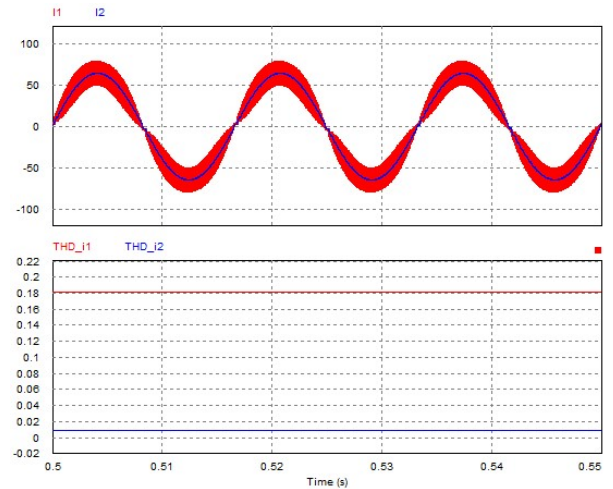


Fig. 8. Waveform of I_1 , I_2 , $THDi1$ and $THDi2$

In Fig. 8, FFT result of I_2 is illustrated for spectrum analysis, and $THDi2$ considering the switching ripples and the

first side bands frequency components is calculated as 0.0027. Fig. 9 shows waveform of converter side current I_1 , grid side current I_2 , THD_{i1} and THD_{i2} in the steady state, and design targeted THD_{i1} , THD_{i2} and current ripple attenuation shows good agreement with the simulation result values. The waveforms show that current control works in a stable manner with the designed LCL filter. The simulation results show that the designed LCL filter is well applied to the inverter and the current is controlled well.

V. CONCLUSION

This paper has proposed a new design method of LCL filter for single phase grid connected power converter in the railway application. LCL filter has been designed by considering total harmonic distortion, reactive power and minimization of magnetic materials. Converter side filter inductor has considered the ripple factor of the switching frequency component to the rated current and the filter capacitor has been limited by the amount of reactive power absorbed under the rated grid voltage condition. The grid side filter inductor has been designed by considering the attenuation ratio of the switching frequency component of the converter side current to the grid side current. The validity of the LCL filter design has been verified by confirming the current control stability and current ripple attenuation in the simulation of single phase grid connected converter with the designed LCL filter.

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